

Table 6.5a False Negative Signals

Period	Market Prices		Period Returns		Excess Returns	
	A	B	A	B	A	B
0	\$10.00	\$20.00				
1	\$11.42	\$22.17	13.3%	10.3%	7.0%	5.3%
2	\$11.12	\$25.48	−2.6%	13.9%	−8.8%	8.9%
3	\$12.60	\$28.62	12.5%	11.6%	6.3%	6.6%
4	\$12.96	\$33.56	2.8%	15.9%	−3.4%	10.9%
5	\$16.91	\$30.59	26.6%	−9.3%	20.4%	−14.3%
6	\$17.63	\$33.58	4.2%	9.3%	−2.0%	4.3%
7	\$17.78	\$37.86	0.8%	12.0%	−5.4%	7.0%
8	\$19.93	\$38.93	11.4%	2.8%	5.2%	−2.2%
9	\$23.13	\$38.94	14.9%	0.0%	8.7%	−5.0%
10	\$24.21	\$39.64	4.6%	1.8%	−1.6%	−3.2%
11	\$23.39	\$46.32	−3.5%	15.6%	−9.7%	10.6%
12	\$23.92	\$49.59	2.3%	6.8%	−3.9%	1.8%
13	\$25.50	\$51.45	6.4%	3.7%	0.2%	−1.3%
14	\$23.97	\$56.96	−6.2%	10.2%	−12.4%	5.2%
15	\$27.35	\$56.60	13.2%	−0.6%	7.0%	−5.6%
16	\$31.27	\$57.37	13.4%	1.3%	7.2%	−3.7%
17	\$30.03	\$61.26	−4.0%	6.6%	−10.2%	1.6%
18	\$36.04	\$61.02	18.2%	−0.4%	12.0%	−5.4%
19	\$32.01	\$67.66	−11.9%	10.3%	−18.1%	5.3%
20	\$33.16	\$69.90	3.5%	3.3%	−2.7%	−1.7%
21	\$37.32	\$66.33	11.8%	−5.2%	5.6%	−10.2%
22	\$34.71	\$73.60	−7.3%	10.4%	−13.5%	5.4%
23	\$39.08	\$71.58	11.9%	−2.8%	5.7%	−7.8%
24	\$44.33	\$66.43	12.6%	−7.5%	6.4%	−12.5%
Avg:			6.2%	5.0%	0.0%	0.0%
Correl:			−0.71		−0.71	

Example 2: False positive signal calculations

Table 6.5b contains the data for two stocks C and D that are moving in opposite directions. Figure 6.7c illustrates this movement over 24 periods. But when we calculate the covariance between these stocks we get a negative correlation, $\rho = +0.90$. How can stocks that move in the same direction have a negative covariance term? The answer is due to the excess terms being on the same side of the price trend. Figure 6.7d illustrates the excess return in each time period. Notice that these excess returns are now on opposite sides of the trend which results in a negative covariance measure.